

5 SEMICONDUCTOR IN EQUILIBRIUM

Essentially, we are interested in determining the current-voltage characteristics of semiconductor devices.

- Since current is due to net flow of charges, there is a need to determine the number of charge carriers available for generating the currents, i.e. the electron concentration in the conduction band and hole concentration in the valence band.

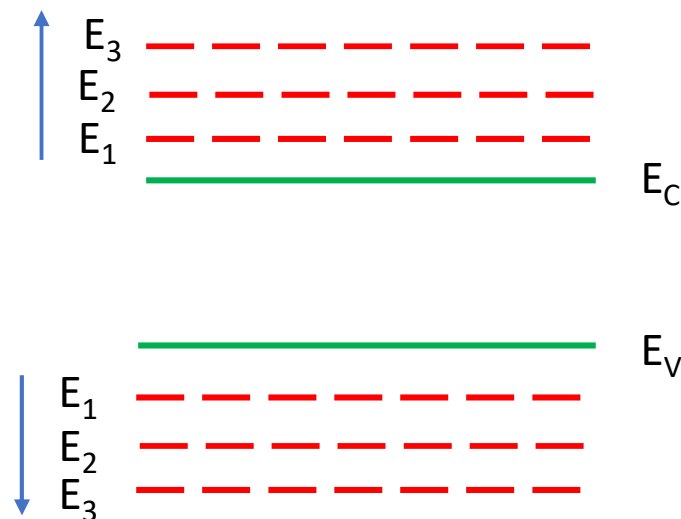
First of all, we will look at the electrons and hole concentrations in thermal equilibrium at temperature T

- Thermal equilibrium implies that no external forces, such as voltages, electric fields, magnetic fields or temperature gradients are acting on the semiconductor
- All properties of semiconductor will be independent of time

- With this starting point, we will then be able to determine the properties of the semiconductor when external excitations are applied

To determine the thermal equilibrium electron and hole concentrations at temperature T , essentially we need to know the following:

- The density of quantum states, that is the number of quantum states per unit volume of the crystal per unit energy, in the conduction and valence bands
- The probability of a quantum states at energy E is occupied by an electron at temperature T



5.1 Fermi-Dirac Distribution Function

Let us, first of all, look at the probability that a quantum state is occupied by an electron at temperature T . This is given by the Fermi-Dirac distribution function.

The Fermi-Dirac probability function describes the distribution of electrons over a range of allowed energy levels at thermal equilibrium. It is based on the following assumptions:

- particles are indistinguishable
- only one particle is permitted in each quantum state (Pauli's exclusion principle)
- The probability of occupancy of each quantum state is the same

- The Fermi-Dirac probability function is given by

$$f(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{k_B T}\right]} \quad (5.1) \quad \text{可能性}$$

- it gives the probability that a quantum state at energy E will be occupied by an electron at temperature T
- it is also the ratio of the filled states to the total quantum states at energy E

Here, the parameter E_F is called the Fermi energy level. It determines the statistical distribution of electrons and does not have to correspond to an allowed energy state. The Fermi level is just a reference level, but very important reference level.

At $T = 0$ K,

$$f(E) = \begin{cases} 1 & \text{for } E < E_F \\ 0 & \text{for } E > E_F \end{cases} \quad (5.2)$$

- all electrons are in their lowest possible energy levels, up to the Fermi level
- probability of quantum states being occupied is unity for $E < E_F$

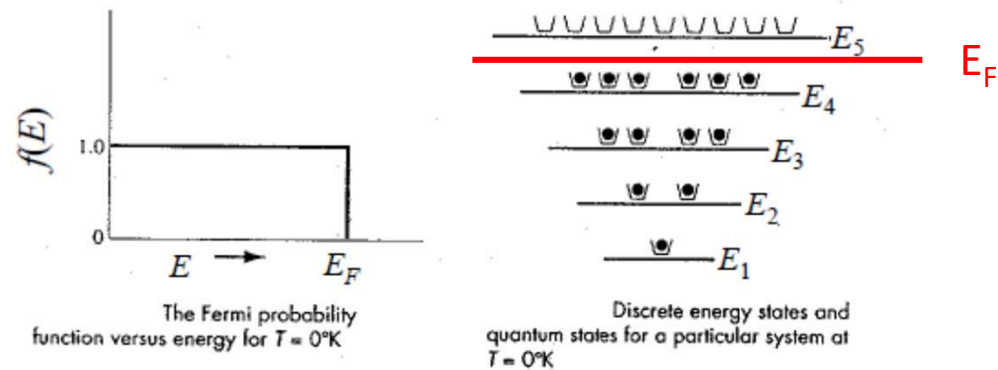


Fig. 5.1

At $T > 0$ K,

- electrons gain a thermal energy so that some of them can jump to higher energy levels
 - change in distribution of electrons among available energy states

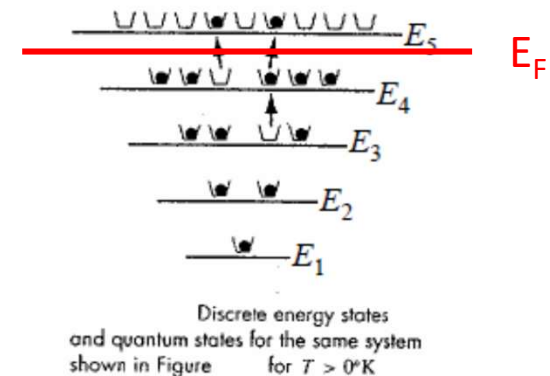


Fig. 5.2

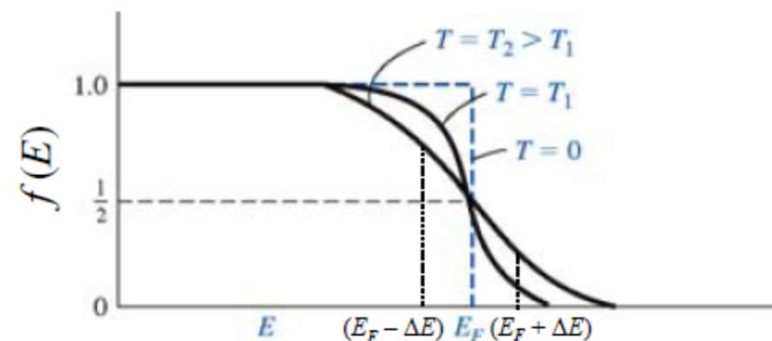
- At $E > E_F$, the Fermi-Dirac distribution in eqn. (5.1) is no longer zero. This means that some probabilities exist for a state above E_F is occupied by an electron.
- At $E < E_F$, the Fermi-Dirac distribution in eqn. (5.1) is no longer unity. This means that some probabilities exist for a state below E_F is empty.

- At $E = E_F$, the Fermi-Dirac distribution is

$$f(E = E_F) = \frac{1}{1 + \exp(0)} = \frac{1}{2} \quad (5.3)$$

i.e. the probability of a state having energy E equal to E_F being occupied an electron is 1/2

- Fig. 5.3 shows the Fermi-Dirac distribution function at various temperature T
 - The Fermi-Dirac function is symmetrical about the Fermi energy E_F at all temperature.



The Fermi probability function vs energy for different temperatures

Fig. 5.3

- Consider the Fermi-Dirac function at temperature T_2 in Fig. 5.3
 - The probability of a state having energy ΔE above E_F , ($E_F + \Delta E$), to be occupied by an electron is $f(E_F + \Delta E)$
 - The probability of a state having energy ΔE below E_F , ($E_F - \Delta E$), to be empty is $[1 - f(E_F - \Delta E)]$
 - Since the Fermi-Dirac function is symmetrical about the Fermi energy E_F , it follows that the probability of a state having energy ΔE above E_F is occupied is the same as the probability of a state having energy ΔE below E_F is empty.
 - mathematically, this means (and can be easily proved)

$$\underbrace{f(E_F + \Delta E)}_{\substack{\text{probability of a state} \\ \text{having energy } \Delta E \text{ above } E_F \\ \text{is occupied}}} = \underbrace{1 - f(E_F - \Delta E)}_{\substack{\text{probability of a state} \\ \text{having energy } \Delta E \text{ below } E_F \\ \text{is empty}}} \quad (5.4)$$

Empty = occupied by hole

- In terms of electrons and holes, we can say that probability of not finding an electron at energy E , $[1 - f(E)]$, is the probability of finding a hole at energy E .

Example 5.1

Calculate the probability that an energy state 0.56 eV above the Fermi energy is occupied by an electron for temperatures of 0 °C and 100 °C.

The probability of an energy state $E = E_F + 0.56 \text{ eV}$ to be occupied by an electron is

$$f(E) = \frac{1}{1 + \exp \left[\frac{E - E_F}{k_B T} \right]} = \frac{1}{1 + \exp \left[\frac{0.56 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} T} \right]}$$

Hence, for $T = 0 \text{ °C} = 273 \text{ K}$, $f(E) = 4.69 \times 10^{-11}$,

and for $T = 100 \text{ °C} = 373 \text{ K}$, $f(E) = 2.76 \times 10^{-8}$

Note that as temperature increases by 100 °C, the probability increases by ~ 600 times. This shows the great sensitivity of this probability on temperature

5.2 Density of States

Having understood the Fermi-Dirac distribution function, we now need to consider the density of quantum states available for electrons to occupy.

5.2.1 Free electrons

To discuss the density of states of electrons in a crystal, let us start with the density of states of free electrons in 3-dimensions (3-D).

- Density of states is the number of quantum states per unit volume of the crystal per unit energy.
- The density of states $g(E)$ for free electrons in 3-D is derived in Appendix B. It is given by

$$g(E) = \frac{4\pi(2m)^{3/2}}{h^3} \sqrt{E} \quad (5.5)$$

where the energy E is the free electron energy in 3-D [see Appendix B, eqn. (B.3)], that is

$$E = \frac{\hbar^2 k^2}{2m}$$

so, alternatively we can write eqn. 5.5 as

$$g(E) = \frac{4\pi(2m)^{3/2}}{h^3} \sqrt{\frac{\hbar^2 k^2}{2m}} \quad (5.6)$$

5.2.2 Electrons in crystals

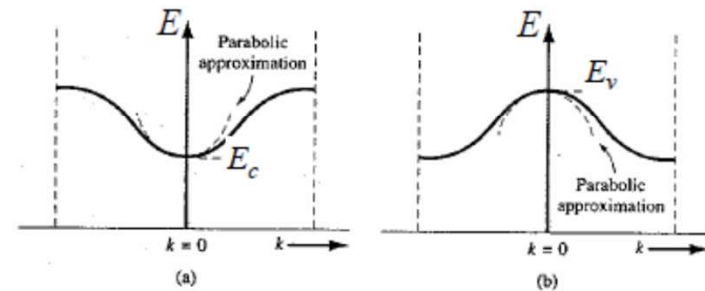
The electrons in a crystal are not completely free, they interact with the periodic potential formed by the ions and electrons in the crystal lattice.

In section 3.4.4, we have seen that the effective mass takes into account the interaction of charge carriers with the lattice, which then allow us to regard the charge carriers as free carriers.

Consider now electrons in the conduction band

- Near the conduction band minima the $E - k$ relation is approximately parabolic (see Fig. 5.4). The electrons behave like free electrons with effective mass m_n^* . Therefore we can express the energy E as

$$E - E_c = \frac{\hbar^2 k^2}{2m_n^*} \quad (5.7)$$



(a) The conduction band in reduced k -space, and the parabolic approximation
(b) The valence band in reduced k -space, and the parabolic approximation

Fig.5.4

- Using eqn. (5.7) and the effective mass m_n^* in eqn. (5.6), we obtain the density of states in the conduction band $g_c(E)$

$$g_c(E) = \frac{4\pi (2m_n^*)^{3/2}}{h^3} \sqrt{E - E_c} \quad (5.8)$$

We can obtain the density of states in the valence band $g_v(E)$ in the same way as we obtain the density of states in the conduction band

- Near the top of the valence band, the $E - k$ relation is also approximately parabolic (see Fig. 5.4) with the effective mass m_p^* . The energy E is then

$$E_v - E = \frac{\hbar^2 k^2}{2m_p^*} \quad (5.9)$$

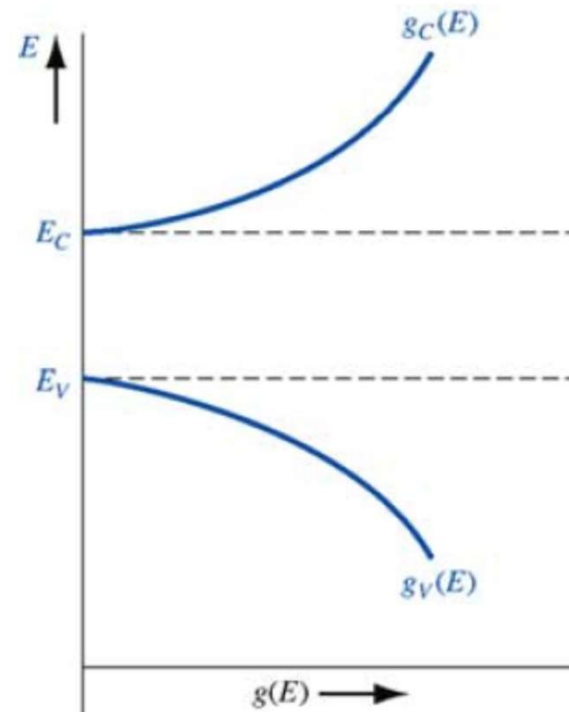
- Therefore, the density of states in the valence band is

$$g_v(E) = \frac{4\pi (2m_p^*)^{3/2}}{h^3} \sqrt{E_v - E} \quad (5.10)$$

Fig. 5.5 shows the density of states in the conduction and valence bands as function of energy E

If quantum states do not exist within forbidden energy band, the density of states

$$g(E) = 0 \text{ for } E_v < E < E_c$$



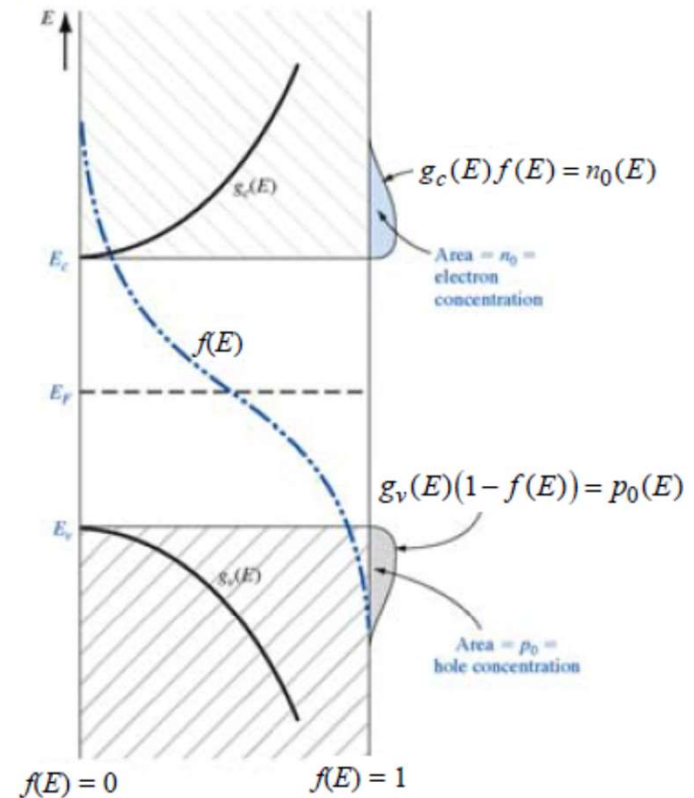
The density of states in the conduction and valence bands as function of energy E

Fig. 5.5

5.3 Thermal equilibrium carrier concentrations

- At thermal equilibrium, the electron concentration at energy E in the conduction band, denoted by $n_0(E)$, is the product of density of the states at energy E in the conduction band and the probability that a state at energy E is occupied by an electron (see Fig. 5.6), that is

$$n_0(E) = g_c(E) f(E) \quad (5.11)$$



Density of states function, Fermi-Dirac probability function, and areas representing electron and hole concentration for the case E_F is near the midgap

Fig. 5.6

- The thermal equilibrium electron concentration in the conduction band n_0 is then given by the integration over all the energies E in the entire conduction band, i.e. from the bottom of the conduction band E_c to ∞

$$n_0 = \int_{E_c}^{\infty} n(E) dE = \int_{E_c}^{\infty} g_c(E) f(E) dE \quad (5.12)$$

- We will calculate the thermal equilibrium electron concentration n_0 for the Fermi energy E_F within the bandgap below the conduction band edge E_c , such that

$$E_c - E_F \gg k_B T \quad (5.13)$$

To satisfy eqn. (5.13), it is sufficient to take $E_c - E_F > 3k_B T$

- The energy of electrons in the conduction band E is above E_F , it follows

$$E - E_F \gg k_B T \quad (5.14)$$

The Fermi-Dirac distribution can then be reduced as follow

$$f(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{k_B T}\right]} \approx \exp\left[-\frac{E - E_F}{k_B T}\right] \quad (5.15)$$

- Now, we evaluate the thermal equilibrium electron concentration n_0 in eqn. (5.12) using the density of states $g_c(E)$ in eqn. (5.8) and the reduced $f(E)$ in eqn. (5.15). We obtain

$$n_0 = N_c \exp \left[-\frac{E_c - E_F}{k_B T} \right] \quad (5.16)$$

where N_c is the effective density of states in the conduction band, given by

$$N_c = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2} \quad (5.17)$$

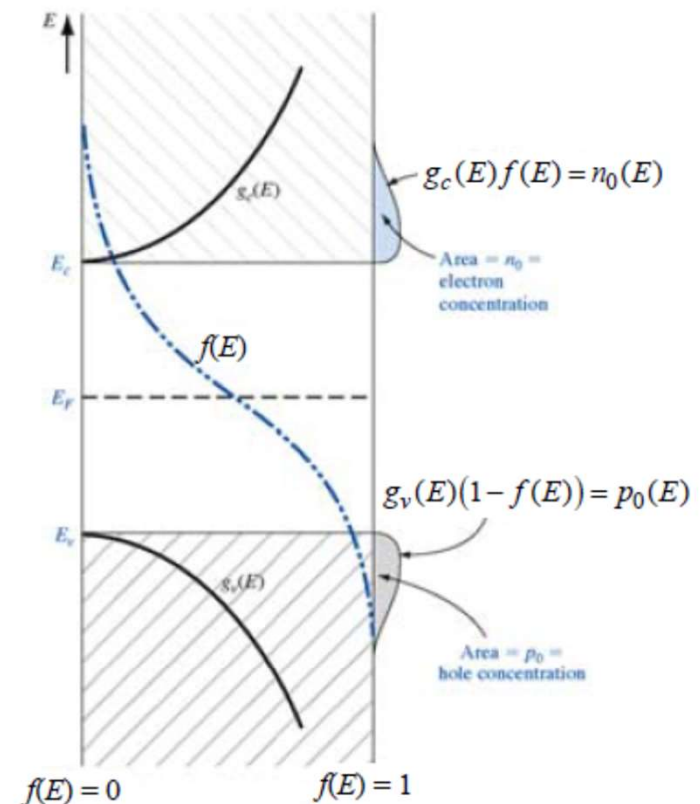
$2.8 \times 10^{19} \text{ cm}^{-3}$ at 300 K

In a similar way, we can obtain the thermal equilibrium hole concentration in the valence band, denoted by p_0 .

- The thermal equilibrium hole concentration at energy E in the valence band, $p_0(E)$, is the product of the density of states at energy E in the valence band and the probability of a state at energy E is not occupied by an electron (see Fig. 5.6), that is

$$p_0(E) = g_v(E) \{1 - f(E)\} \quad (5.18)$$

$\swarrow$
 the probability of a state at
 energy E is not occupied
 by an electron



Density of states function, Fermi-Dirac probability function, and areas representing electron and hole concentration for the case E_F is near the midgap

Fig. 5.6

- The thermal equilibrium holes concentration in the valence band is then given by

$$p_0 = \int_{-\infty}^{E_v} g_v(E) \{1 - f(E)\} dE \quad (5.19)$$

the integration is over all the energy E in the valence band, i.e. from $-\infty$ to the top of valence band E_v .

- Now, we evaluate p_0 for the Fermi energy E_F above the valence band edge E_v , such that

$$E_F - E_v \gg k_B T \quad (5.20)$$

Again, $E_F - E_v > 3k_B T$ is sufficient to satisfy eqn. (5.20)

- Since the energy E in the valence band is below E_v , it follows

$$E_F - E \gg k_B T \quad (5.21)$$

Using eqn. (5.21), the term $\{1 - f(E)\}$ in eqn. (5.19) can then be approximated as follow

$$1 - f(E) = 1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} \approx \exp\left(-\frac{E_F - E}{k_B T}\right) \quad (5.22)$$

- We can now evaluate the thermal equilibrium hole concentration p_0 in eqn. (5.19) using the density of states $g_v(E)$ in eqn. (5.10) and the reduced $\{1 - f(E)\}$ in eqn. (5.22). We obtain

$$p_0 = N_v \exp \left[- \frac{E_F - E_v}{k_B T} \right] \quad (5.23)$$

where N_v is the effective density of states in valence band and given by

$$N_v = 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2} \quad (5.24)$$

$1.04 \times 10^{19} \text{ cm}^{-3}$ at 300K

It is important to note the following

- The assumption used in deriving the thermal equilibrium electron and hole concentrations is that the Fermi energy E_F is not too close to the conduction and valence band edges, E_c and E_v respectively. So long as this assumption is valid, the expressions for the thermal equilibrium electron and hole concentrations, n_0 and p_0 , are valid for intrinsic and extrinsic semiconductors.
- n_0 and p_0 are directly related to the effective density of states and to the Fermi energy level
 - temperature factor present in the effective density of states and the exponential term (representative of probability)
 - hence, thermal equilibrium concentrations will be very sensitive to the temperature

5.4 Intrinsic Material

For an intrinsic semiconductor,

- concentration of electrons in conduction band is equal to concentration of holes in valence band
- if we denote the electron and hole concentrations in an intrinsic semiconductor as n_i and p_i (subscript i refers to intrinsic), then

$$\boxed{n_i = p_i} \quad (5.25)$$

- the Fermi energy level of an intrinsic semiconductor is called the intrinsic Fermi energy, E_i . We understand that the expression of the thermal equilibrium electron and hole concentrations, eqns. (5.16) and (5.23) respectively, valid for both intrinsic and extrinsic semiconductors. If we apply eqns. (5.16) and (5.23) to an intrinsic semiconductor we obtain

$$n_i = N_c \exp\left[-\frac{E_c - E_i}{k_B T}\right] \quad (5.26)$$

and

$$p_i = N_v \exp\left[-\frac{E_i - E_v}{k_B T}\right] \quad (5.27)$$

- Now, multiply eqn. (5.26) with eqn. (5.27) and use the condition in eqn. (5.25) we obtain

$$n_i^2 = N_c N_v \exp\left[-\frac{E_c - E_v}{k_B T}\right] = N_c N_v \exp\left[-\frac{E_g}{k_B T}\right] \quad (5.28)$$

$$n_i = \sqrt{N_c N_v} \exp\left[-\frac{E_g}{2k_B T}\right] \quad (5.29)$$

It is very important to note here, that for a given semiconductor material at a constant temperature, the value of the intrinsic carrier concentration, n_i , is a constant and is independent of the Fermi energy. This is true regardless whether the semiconductor is intrinsic (undoped) or extrinsic (doped). We can always calculate the intrinsic carrier concentration, n_i , using equation (5.29).

- Recall eqns. (5.16) and (5.23), the thermal equilibrium electron and hole concentrations, n_0 and p_0 . The product of n_0 and p_0 is

$$n_0 p_0 = N_c N_v \exp\left[-\frac{E_c - E_v}{k_B T}\right] = N_c N_v \exp\left[-\frac{E_g}{k_B T}\right] \quad (5.30)$$

We see that the product of n_0 and p_0 is a constant for a particular material and temperature T . This is exactly the same as the expression for n_i^2 in equation (5.28). We, therefore, obtain

$$n_0 p_0 = n_i^2 \quad (5.31)$$

This is an important equation and we will use it extensively later. Note also, that in deriving this equation we did not make any assumption whether the semiconductor is intrinsic or extrinsic. Therefore, this equation valid for intrinsic and extrinsic semiconductors.

- At a particular temperature T , the value of n_i is dependent on material parameters, like electron and hole effective masses, m_n^* and m_p^* respectively, and the bandgap energy E_g .

- For example, silicon at 300 K

$m_n^* = 1.08 m_0$ and $m_p^* = 0.56 m_0$, where m_0 is the free electron rest mass. $E_g = 1.12$ eV.

Using equations (5.17) and (5.24),

$$N_c = 2.8 \times 10^{19} / \text{cm}^3 \text{ and } N_v = 1.05 \times 10^{19} / \text{cm}^3$$

and then using equation (5.29)

$$n_i = 6.84 \times 10^9 / \text{cm}^3$$

- experimentally, the accepted value of n_i for silicon at 300 K is $1.5 \times 10^{10} / \text{cm}^3$. The discrepancy is due to the simple theoretical model adopted.

- Table 5.1 list the value of n_i for some semiconductors at 300 K. We see that n_i increases with decreasing energy gap E_g . This is expected since n_i is exponentially dependent on E_g (see eqn. 5.29)

Table 5.1

Materials	n_i (/cm ³)	E_g (eV)
GaAs	1.8×10^6	1.43
Si	1.5×10^{10}	1.11
Ge	2.4×10^{13}	0.67

- If we substitute the effective density of states N_c in eqn. (5.17) and N_v in eqn. (5.24) into the intrinsic carrier concentration n_i in eqn. (5.29), we obtain

$$n_i = \underbrace{2 \left(\frac{2\pi k_B T}{h^2} \right)^{3/2}}_{\text{relatively constant over temperature range}} \underbrace{(m_n^* m_p^*)^{3/4} \exp \left[-\frac{E_g}{2k_B T} \right]}_{\text{dominant term}} \quad (5.32)$$

- In eqn. (5.32), we note that, for a given material, all the parameters, except T , are constant
 - temperature T appears twice in eqn. (5.32): $T^{3/2}$ and $\exp(-1/T)$.
 - the dominant term is the exponential term, the $T^{3/2}$ is relatively constant over a temperature range.

Hence, we could write

$$n_i \approx \text{constant} \times \exp\left[-\frac{E_g}{2k_B T}\right]$$

$$\ln n_i = \ln(\text{constant}) - \frac{E_g}{2k_B T} \quad (5.33)$$

this is a straight line equation, so if we plot $\ln n_i$ versus $1/T$, we will get a straight line with the slope of $-E_g/2k_B$

- The plot of eqn. (5.33) for some semiconductors is shown in Fig. 5.7.

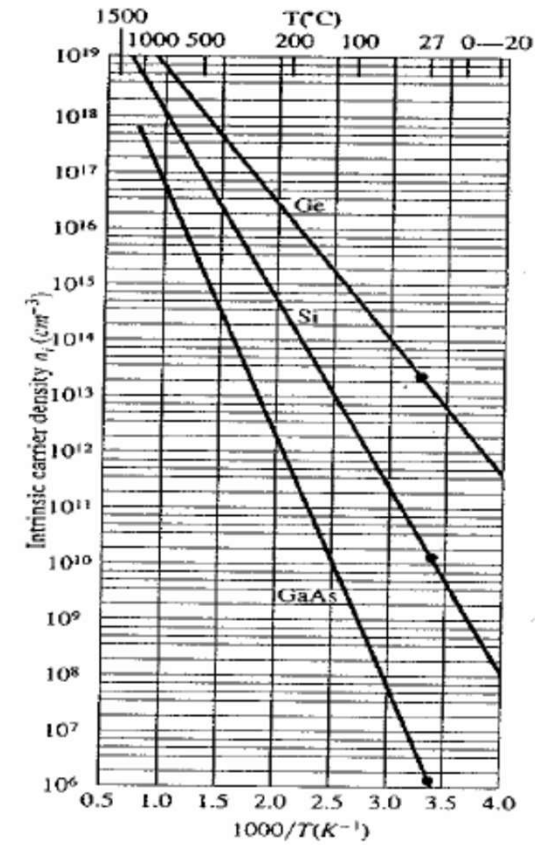


Fig. 5.7

- To calculate the exact position of the intrinsic Fermi level E_i , we equate n_i in eqn. (5.26) and p_i in eqn. (5.27). We obtain

$$E_i = \frac{1}{2}(E_c + E_v) + \frac{1}{2}k_B T \ln\left(\frac{N_v}{N_c}\right) \quad (5.34)$$

Using the expressions of N_c and N_v in eqns. (5.17) and (5.24) respectively, we change the ratio of the effective density of states into the ratio of the effective masses

$$E_i = E_{mid} + \frac{3}{4}k_B T \ln\left(\frac{m_p^*}{m_n^*}\right) \quad (5.35)$$

where $E_{mid} = \frac{1}{2}(E_c + E_v)$ (5.36)

- From eqn. (5.36), we see that the intrinsic Fermi level is exactly in the middle of the bandgap, i.e. $E_i = E_{mid}$, if
 - the temperature $T = 0$ K or
 - $m_n^* = m_p^*$
- The deviation of E_i from E_{mid} is very small and in many cases the position E_i can be considered to be at the center of the bandgap.
- Knowing the material parameters (m_n^* , m_p^* and E_g), we can always calculate the intrinsic carrier concentration n_i and the intrinsic Fermi level E_i at any temperature T using eqns. (5.32) and (5.35) respectively. This is true whether the semiconductor is intrinsic or extrinsic.

- It is very convenient and useful to express the thermal equilibrium electron and holes concentrations, n_0 and p_0 , in terms of the intrinsic carrier concentration n_i and the intrinsic Fermi level E_i . We can easily do that by dividing eqn. (5.16) with eqn. (5.26) and eqn. (5.23) with eqn. (5.27). We obtain

$$n_0 = n_i \exp \left[\frac{E_F - E_i}{k_B T} \right] \quad (5.37)$$

$$p_0 = n_i \exp \left[\frac{E_i - E_F}{k_B T} \right] \quad (5.38)$$

These eqns. are very useful and important, we will use it extensively later. Multiplying eqn. (5.37) with eqn. (5.38) will give us

$$n_0 p_0 = n_i^2$$

which is exactly the same result as we obtained in eqn. (5.31).

5.5 Extrinsic materials

In Chapter 4 we have learnt how the donor impurity donates electrons to the conduction band (forming n-type material) and the acceptor impurity creates holes in the valence band (forming p-type material).

In this section we will calculate the thermal equilibrium electron and hole concentrations, n_0 and p_0 , as a function of the impurity concentration.

5.5.1 Donor impurity

Let us start with an n-type semiconductor where donor concentration is N_d . The majority carrier is electrons.

From section 4.2.:

- The thermal equilibrium majority electron concentration n_0 is the sum of the electron concentration from the donor impurity and the electron concentration due to the excitation from the valence band to the conduction band creating the electron-hole pairs.

- At temperature T where all donor atoms are all ionized we have (recall eqn. 4.1)

$$n_0 = N_d + p_0 \quad (4.1)$$

negative charge concentration positive charge concentration

- How can we now calculate n_0 and p_0 ?

We first calculate the majority carrier concentration n_0 by simply using eqn. (5.31)

$$n_0 p_0 = n_i^2$$

to substitute p_0 in eqn. (4.1) in terms of n_0 and n_i .

We obtain

$$n_0 = N_d + \frac{n_i^2}{n_0} \quad (5.39)$$

Multiplying eqn. (5.39) throughout by n_0 and rearranging the equation, we get

$$n_0^2 - N_d n_0 - n_i^2 = 0 \quad (5.40)$$

this is a quadratic equation in n_0 . The solution of this equation is

$$n_0 = \frac{N_d \pm \sqrt{N_d^2 + 4n_i^2}}{2} \quad (5.41)$$

- We only take the positive solution in eqn. (5.41). The negative solution is omitted since it has no physical meaning.

- So, the thermal equilibrium majority electron concentration is

$$n_0 = \frac{N_d + \sqrt{N_d^2 + 4n_i^2}}{2} \quad (5.42)$$

- It is important to note the following in eqn. (5.42):
 - If the doping concentration is much larger than the intrinsic carrier concentration, that is

$$N_d \gg n_i \quad (5.43)$$

then the $4n_i^2$ term in eqn. (5.42) can be ignored and n_0 is approximately equal to N_d ,

$$n_0 \approx N_d \quad (5.44)$$

- If the doping concentration is not much larger than the intrinsic carrier concentration, then n_0 cannot be approximated to N_d , and eqn. (5.42) needs to be used to calculate n_0 .
- Once we obtain n_0 , the thermal equilibrium minority hole concentration can be calculated using eqn. (5.31)

$$p_0 n_0 = n_i^2$$

- Using the value of n_0 , the position of the Fermi level E_F in the bandgap can be determined using either eqn. (5.16) or eqn. (5.37), depending on the question, of course

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{k_B T}\right] \quad (5.16)$$

$$n_0 = n_i \exp\left[\frac{E_F - E_i}{k_B T}\right] \quad (5.37)$$

- At thermal equilibrium, the position of the Fermi level in the bandgap can also be determined using the value of p_0 . In this case, we use either eqn. (5.23) or eqn. (5.38).

$$p_0 = N_v \exp \left[-\frac{E_F - E_v}{k_B T} \right] \quad (5.23)$$

$$p_0 = n_i \exp \left[\frac{E_i - E_F}{k_B T} \right] \quad (5.38)$$

At thermal equilibrium, the position of the Fermi level E_F is the same regardless we use n_0 or p_0 (see example 5.2)

- For n-type semiconductor ($n_0 > p_0$), E_F is above the intrinsic Fermi level E_i . The Fermi level E_F moves up as n_0 increases. This is clear from eqns. (5.16) and (5.37) and illustrated in Fig. 5.8.

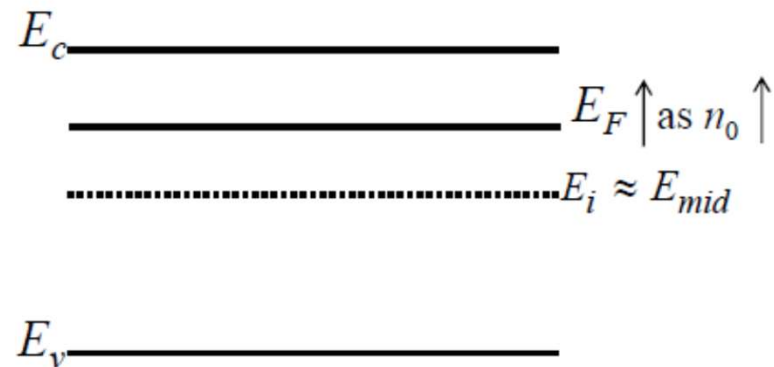


Fig. 5.8

Example 5.2

Consider a Si sample doped with 10^{16} P atoms/cm³ at 300 K. The intrinsic carrier concentration is 1.5×10^{10} /cm³. Calculate

- i) the thermal equilibrium minority concentration
- ii) and the position of the Fermi level relative to the intrinsic Fermi level

The impurity P atom is from Group V in the periodic table. So, in the Si sample, the P atom acts as donor and hence the majority carrier is electrons.

Using eqn. (5.42) we obtain the majority carrier concentration

$$n_0 = \frac{10^{16} + \sqrt{(10^{16})^2 + 4(1.5 \times 10^{10})^2}}{2} = 10^{16} / \text{cm}^3$$

From the above calculation we show that for $N_d \gg n_i$, $n_0 = N_d$

From eqn. (5.31), the thermal equilibrium minority hole concentration is

$$n_0 p_0 = n_i^2$$
$$p_0 = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4 \text{ /cm}^3$$

The position the Fermi level can be obtained using either the value of n_0 (eqn. 5.37) or p_0 eqn. (eqn. 5.38)

Using the value of n_0

$$n_0 = n_i \exp \left[\frac{E_F - E_i}{k_B T} \right] \quad (5.37)$$

$$E_F - E_i = k_B T \ln \left(\frac{n_0}{n_i} \right) = 1.38 \times 10^{-23} \times 300 \times \ln \left(\frac{10^{16}}{1.5 \times 10^{10}} \right) = 5.55 \times 10^{-20} \text{ J}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, \quad \therefore E_F - E_i = 0.347 \text{ eV}$$

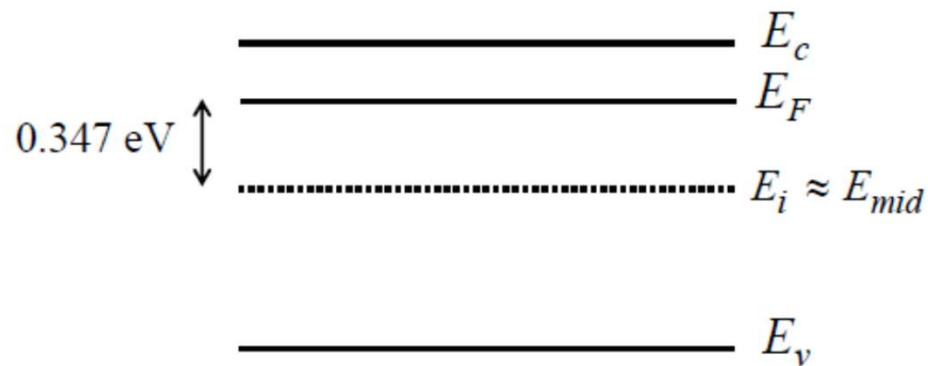
Using the value of p_0

$$p_0 = n_i \exp\left[\frac{E_i - E_F}{k_B T}\right] \quad (5.38)$$

$$E_i - E_F = k_B T \ln\left(\frac{p_0}{n_i}\right) = 1.38 \times 10^{-23} \times 300 \times \ln\left(\frac{2.25 \times 10^4}{1.5 \times 10^{10}}\right) = -5.55 \times 10^{-20} \text{ J}$$

$$E_i - E_F = -0.347 \text{ eV} \text{ or } E_F - E_i = 0.347 \text{ eV}$$

This is exactly the same result as obtained previously using eqn. (5.37)



Example 5.3

Consider a germanium sample at 330 K in which $N_d = 5 \times 10^{13} \text{ /cm}^3$. Assume $n_i = 2.4 \times 10^{13} \text{ /cm}^3$, calculate the thermal equilibrium majority and minority concentration.

At 330 K, the donor impurity is fully ionized. From eqn. (5.42), the majority carrier concentration is

$$n_0 = \frac{5 \times 10^{13} + \sqrt{(5 \times 10^{13})^2 + 4(2.4 \times 10^{13})^2}}{2} = 5.97 \times 10^{13} \text{ /cm}^3$$

Notice that, since N_d is not much larger than n_i , $n_0 \neq N_d$

From eqn. (5.31), the minority carrier concentration is

$$p_0 = \frac{n_i^2}{n_0} = \frac{(2.4 \times 10^{13})^2}{5.97 \times 10^{13}} = 9.65 \times 10^{12} \text{ /cm}^3$$

5.5.2 Acceptor impurity

Let us now consider a p-type semiconductor with acceptor concentration N_a . Here, the majority carrier is holes.

From section 4.3:

- The thermal equilibrium majority hole concentration p_0 is the sum of the holes created due to the excitation of electrons from the valence band to the acceptor level and the excitation of electron from the valence band to the conduction band.
- At T where all acceptor atoms are fully ionized we have (see eqn. 4.6)

$$p_0 = N_a + n_0 \quad (4.6)$$

positive charge concentration

negative charge concentration

- We can obtain the majority carrier concentration p_0 and the minority carrier concentration n_0 by using the same procedure as we did in section 5.5.1.
- Firstly, we calculate the majority carrier concentration p_0 by substituting the minority carrier concentration n_0 in eqn. (4.6) in terms of p_0 and n_i . Of course, we use eqn. (5.31). We obtain

$$p_0 = N_a + \frac{n_i^2}{p_0} \quad (5.45)$$

from eqn. (5.45), the majority carrier concentration is

$$p_0 = \frac{N_a + \sqrt{N_a^2 + 4n_i^2}}{2} \quad (5.46)$$

- Again, using eqn. (5.31), the minority carrier concentration is

$$n_0 = \frac{n_i^2}{p_0}$$

- For $N_a \gg n_i$, $p_0 \approx N_a$ (5.47)
- Having obtained p_0 and n_0 , the position of the Fermi level can be easily determined as discussed in section 5.5.1.
 - Using the value of p_0 we use

$$\text{either } p_0 = N_v \exp \left[-\frac{E_F - E_v}{k_B T} \right] \quad (5.23)$$

$$\text{or } p_0 = n_i \exp \left[\frac{E_i - E_F}{k_B T} \right] \quad (5.38)$$

- Using the value of n_0 we use

$$\text{either } n_0 = N_c \exp\left[-\frac{E_c - E_F}{k_B T}\right] \quad (5.16)$$

$$\text{or } n_0 = n_i \exp\left[\frac{E_F - E_i}{k_B T}\right] \quad (5.37)$$

- For p-type semiconductor ($p_0 > n_0$), E_F is below E_i . The Fermi level E_F moves down with increasing p_0 . This is clear from eqns. (5.23) and (5.38) and illustrated in Fig. 5.9.



Fig. 5.9

5.5.3 Compensated semiconductor

A compensated semiconductor is a semiconductor that contains both donor and acceptor impurity atoms in the same region.

- It can be formed by diffusing acceptor impurities into n-type material or donor impurities into a p-type material
- Considering all impurities are fully ionized
 - n-type is created when $N_d > N_a$
 - p-type is created when $N_a > N_d$
 - Intrinsic is created when $N_d = N_a$

- Compensated semiconductors are created naturally during device fabrication.

Fig. 5.10 shows the formation a p-type semiconductor by diffusing acceptor boron impurities into an n-type Si sample thereby forming a p-n junction.

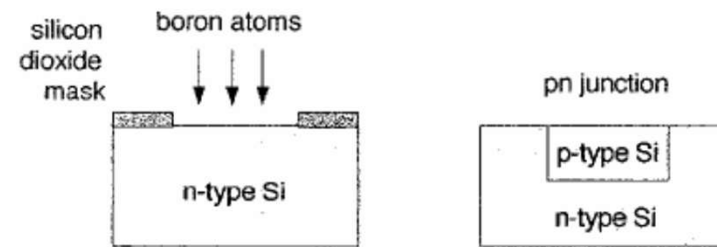


Fig. 5.10

- We will now calculate the thermal equilibrium carrier concentration, n_0 and p_0 , where both donor and acceptor impurities are present.

Fig. 5.11 shows the electron transitions among the energy levels and bands (E_d , E_a , E_c and E_v) forming thermal equilibrium carrier concentrations. How are we going to determine n_0 and p_0 ?

- At thermal equilibrium the semiconductor is electrically neutral. As we did in previous sections (sections 5.5.1 and 5.5.2) we will use the charge neutrality condition to obtain the thermal equilibrium carrier concentrations.

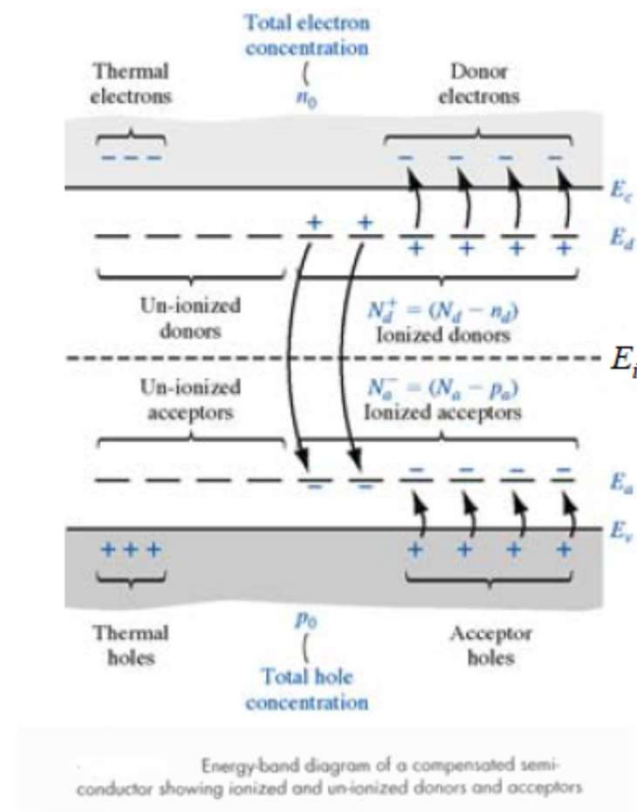


Fig. 5.11

- The charge neutrality (often referred to as the space charge neutrality):
The sum of positive charges = the sum of negative charges
- The general form of the charge neutrality condition is

$$\underbrace{p_0 + N_d^+}_{\text{sum of positive charges}} = \underbrace{n_0 + N_a^-}_{\text{sum of negative charges}} \quad (5.48)$$

Here, for generality, we have considered a relatively low temperature where the impurities are not fully ionized. So, we have in eqn. (5.48)

- N_d^+ : ionized donor concentration
- N_a^- : ionized acceptor concentration

- At temperature T where all impurities are ionized eqn. (5.48) becomes

$$\boxed{p_0 + N_d = n_0 + N_a} \quad (5.49)$$

Note that, when only donor impurity is present, i.e. $N_a = 0$, eqn. (5.49) reduces to eqn. (4.1)

$$p_0 + N_d = n_0$$

and when only acceptor impurity is present, i.e. $N_d = 0$, eqn. (5.49) reduces to eqn. (4.6)

$$p_0 = n_0 + N_a$$

- Let us calculate the thermal equilibrium carrier concentration where both donor and acceptor impurities are present and consider only the case where the temperature is high enough so that all impurities are ionized.

We start with eqn. (5.49): $p_0 + N_d = n_0 + N_a$

- Consider first a case where $N_d > N_a$, i.e. n-type semiconductor. As before, we express p_0 in eqn. (5.49) in terms of n_0 and n_i using eqn. (5.31), we obtain

$$n_0^2 - (N_d - N_a)n_0 - n_i^2 = 0 \quad (5.50)$$

The solution this quadratic equation is

$$n_0 = \frac{(N_d - N_a) + \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2} \quad (5.51)$$

as usual we obtain the minority carrier concentration p_0 from eqn. (5.31)

- For $N_a > N_d$, i.e. p-type semiconductor, we express n_0 in terms of p_0 and n_i . We can then write eqn. (5.49) as follow

$$p_0^2 - (N_a - N_d)p_0 - n_i^2 = 0 \quad (5.51)$$

The solution is

$$p_0 = \frac{(N_a - N_d) + \sqrt{(N_a - N_d)^2 + 4n_i^2}}{2} \quad (5.52)$$

the minority carrier concentration n_0 is then obtained from eqn. (5.31)

- It is important to note here that the charge neutrality equation is always true and can be used to determine the carrier concentrations in a semiconductor sample regardless
 - whether there is only one type of impurity or two types of impurities present
 - and whether the concentrations of the impurities are much larger or in the order of n_i
- From eqns. (5.42), (5.46), (5.51) or (5.52), we see that the majority carrier concentration is not a simple sum of dopant concentration and intrinsic carrier concentration.

Example 5.4

A piece of germanium crystal is doped with donor impurities to a concentration of $5.8 \times 10^{14} \text{ /cm}^3$, as well as acceptor impurities to a concentration of $5.4 \times 10^{14} \text{ /cm}^3$. Taking the intrinsic concentration at 300 K to be $2.4 \times 10^{13} \text{ /cm}^3$, calculate the thermal equilibrium electron and hole concentrations.

Material has 2 types of impurities $N_d = 5.8 \times 10^{14} \text{ /cm}^3$

$$N_a = 5.4 \times 10^{14} \text{ /cm}^3$$

Since $N_d > N_a$, the material is an n-type compensated material.

From eqn. (5.51), the majority carrier concentration is

$$n_0 = \frac{(N_d - N_a) + \sqrt{(N_d - N_a)^2 + 4n_i^2}}{2}$$
$$n_0 = \frac{(5.8 \times 10^{14} - 5.4 \times 10^{14}) + \sqrt{(5.8 \times 10^{14} - 5.4 \times 10^{14})^2 + 4(2.4 \times 10^{13})^2}}{2}$$

$$n_0 = 5.12 \times 10^{13} \text{ /cm}^3$$

From eqn. (5.31), the minority carrier concentration is

$$p_0 = \frac{(2.4 \times 10^{13})^2}{5.12 \times 10^{13}} = 1.125 \times 10^{13} \text{ /cm}^3$$